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[22CSBS105]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

COMPUTER ORGANIZATION & ARCHITECTURE

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 X 2=10**		
1.	**a.**	Discuss the purpose of Program Counter (PC) and Instruction Register (IR).	L2	CO1	**02**
	b.	List down the different types of data representation in digital Computer?	L1	CO2	**02**
	c.	Mention different types of basic computer instructions ..	L1	CO3	**02**
	d.	List the privileged and non-privileged instructions?	L1	CO4	**02**
	e.	Define Cache Memory.	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	List down the special registers in a typical computer? Explain their purposes in detail.	L1	CO1	**05**
		OR			
	b.	Compare and contrast between RISC and CISC instructions.	L2	CO1	**05**
3.	**a.**	Describe in detail about 4- Bit Ripple Carry Adder.	L2	CO2	**05**
		OR			
	b.	Solve the following using 2's complement subtraction **i.** 10110 – 11010 **ii.** 10110-01111	L3	CO2	**05**
4.	**a.**	Briefly Explain about parallel processing	L2	CO3	**05**
		OR			
	b.	Illustrate the concept of basic pipeline with diagram.		CO3	**05**
5.	**a.**	Briefly explain about software interrupt and exceptions	L2	CO4	**05**
		OR			
	b.	Describe in detail about programmed I/O modes of data transfer.	L2	C04	**05**
6.	**a.**	Differentiate SRAM and DRAM.	L2	CO5	**05**
		OR			
	b.	Define locality of reference in memory ? Explain its types.	L2	C05	**05**
7.	**a.**	Compare Hardwired control and Micro programmed control.	L1	CO3	**05**
		OR			
	b.	Briefly explain about Three types of basic instruction formats in digital computer.		CO1	**05**

		PART-C			**6 X 10=60**
8.	**a.**	Explain in detail about instruction execution cycle with neat flow chart.	L2	CO1	**10**
		OR			
	b.	Explain in detail about different types of addressing modes in 8086 microprocessor with suitable examples.	L2	CO1	**10**
9.	**a.**	Solve A = 10111 (-9) x B =10011 (-13) using Booth Multiplication algorithm	L3	CO2	**10**
		OR			
	b.	Solve Using Restoring Division algorithm for the following Dividend =1111 (15) Divisor =0100 (4)	L3	CO2	**10**
10.	**a.**	Explain in Detail about architecture of 8086 Microprocessor with a neat Diagram.	L2	CO3	**10**
		OR			
	b.	Explain in detail about Hardwired Control Technique with the help of the diagram.	L2	CO3	**10**
11.	**a.**	Describe DMA controller? With the help of block diagram explain the DMA transfer.	L2	CO4	**10**
		OR			
	b.	Explain in detail about I/O interface with a neat block diagram	L2	C04	**10**
12.	**a.**	Briefly Explain about the following with a neat diagram **i.** Direct Memory Mapping **ii.** Associative Memory Mapping	L2	CO5	**10**
		OR			
	b.	Explain in detail about 2D & 2.5 D Memory Organization with diagram.	L2	CO5	**10**
13.	**a.**	Using IEEE754 standard, Represent (1259.125)10 in single and double precision formats.	L3	CO2	**10**
		OR			
	b.	Solve the following using Page replacement Algorithm. Consider a reference string: 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. the number of frames in the memory is 3. Find out the number of page faults respective to: **i.** Optimal Page Replacement Algorithm **ii.** FIFO Page Replacement Algorithm **iii.** LRU Page Replacement Algorithm	L3	CO5	**10**